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A SEARCH FOR A DISTANT COMPANION TO THE SUN WITH THE *WIDE-FIELD INFRARED SURVEY EXPLORER**

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ABSTRACT

I have used multi-epoch astrometry from the *Wide-field Infrared Survey Explorer* to perform a search for a distant companion to the Sun via its parallactic motion. I have not found an object of this kind down to $W2 = 14.5$. This limit corresponds to analogs of Saturn and Jupiter at 28,000 and 82,000 AU, respectively, according to models of the Jovian planets by Fortney and coworkers. Models of brown dwarfs by Burrows and coworkers predict fainter fluxes at a given mass for the age of the solar system, producing a closer distance limit of 26,000 AU for a Jupiter-mass brown dwarf. These constraints exclude most combinations of mass and separation at which a solar companion has been suggested to exist by various studies over the years.

Key words: brown dwarfs – infrared: planetary systems – planets and satellites: detection – proper motions – solar neighborhood

Online-only material: color figure, machine-readable table

1. INTRODUCTION

Beyond the orbit of Neptune, the solar system is known to contain multiple populations of small, icy bodies that extend out to ∼100,000 AU (Edgeworth 1949; Oort 1950; Kuiper 1951; Jewitt & Luu 1993). It has been speculated that a gas giant planet or a brown dwarf might also reside in the outer solar system. A variety of observations have been cited over the past century as indirect evidence for such an object. For instance, a possible periodicity in mass extinctions has been interpreted as the orbital period of a distant companion that perturbs comets in the Oort cloud during perihelion ($a \sim 100{,}000$ AU, Davis et al. 1984; Whitmire & Jackson 1984). More recently, an anomaly in the orbits of outer Oort cloud comets has been attributed to a 1–4 $M_{\rm Jup}$ object in a somewhat smaller orbit ($a \sim 10{,}000$–$30{,}000$ AU, Matese et al. 1999; Matese & Whitmire 2011). Some aspects of these scenarios have been questioned, such as the stability of very wide orbits (Torbett & Smoluchowski 1984) and the evidence of an extinction periodicity (Feng & Bailer-Jones 2013). Nevertheless, given that substellar companions have been found at separations of >1000 AU from a few nearby stars (Burgasser et al. 2000; Scholz et al. 2003; Luhman et al. 2011), the presence of a similar body in the outer solar system is plausible.

To search for a new member of the solar system, one can seek a detection in either reflected sunlight at optical wavelengths or thermal radiation at infrared (IR) wavelengths. The former has a steeper dependence on distance from the Sun (r^{-4}) but a shallower dependence on mass. As a result, reflected light is generally easier to detect in the inner solar system, unless a body has a low albedo or a large mass (and hence a high temperature). At greater distances from the Sun, thermal radiation dominates for an increasing range of masses until eventually becoming the only option for detection. Jupiter and the largest trans-Neptunian objects (TNOs) span ∼10 mag in the brightness of their reflected light at a given distance, which corresponds to a range of only a factor of ∼10 in the distance limits. For instance, recent reflected light surveys have reached 100–200 AU for TNOs and 1000–2000 AU for gas giants (Trujillo & Brown 2003; Elliot et al. 2005; Larsen et al. 2007; Schwamb et al. 2010; Sheppard et al. 2011). At the outer limit of 2000 AU for those surveys, the *Infrared Astronomical Satellite* (*IRAS*) and the Two Micron All-Sky Survey (2MASS; Skrutskie et al. 2006) were capable of detecting the thermal emission of brown dwarfs with masses as low as ∼3 and 6 $M_{\rm Jup}$, respectively (Burrows et al. 1997). It is likely that the *IRAS* and 2MASS databases have been thoroughly searched for a distant companion to the Sun, although detailed descriptions of such analysis have not been published.

Because of the combination of its mid-IR sensitivity and all-sky coverage, the *Wide-field Infrared Survey Explorer* (*WISE*; Wright et al. 2010) offers the best available constraints on the presence of a large body in the outer solar system, reaching masses that are an order of magnitude lower than the limits from 2MASS and *IRAS*. In addition, the multiple epochs of images collected by *WISE* are perfectly timed for the detection of parallactic motion, providing a discriminating means of identifying a solar companion. This paper presents the results of a search of the *WISE* database for objects exhibiting this signature. It begins with descriptions of the *WISE* mission (Section 2) and my search strategy and data processing (Sections 3 and 4). I then evaluate candidate moving objects (Sections 5 and 6) and characterize the resulting constraints on the presence of a distant companion to the Sun (Section 7).

2. THE *WIDE-FIELD INFRARED SURVEY EXPLORER*

A full description of *WISE* has been provided by Wright et al. (2010). I summarize the aspects of the mission that are relevant to this study.

WISE collected images in four bands that were centered at 3.4, 4.6, 12, and 22 μm, which are denoted as $W1$, $W2$, $W3$, and $W4$. The angular resolution was ∼6″ in the first three bands and ∼12″ in $W4$. The satellite was stationed in a Sun synchronous orbit with a period of 95 minutes. The telescope was pointed at 90° solar elongation, scanning great circles centered on the Sun during its orbit. A $47' \times 47'$ field was observed simultaneously in all operable bands. At intervals of 11 s, one 8.8 s exposure was

* Based on data from the *Wide-field Infrared Survey Explorer* and the Two Micron All-Sky Survey.

obtained in each band at a fixed position on the sky. Adjacent images during an orbit overlapped by 10%. The scans from adjacent orbits overlapped by 90%–100% at ecliptic latitudes of 0°–90°, respectively. As a result, the number of images at a given position and band ranged from 12 near the ecliptic plane to hundreds near the ecliptic poles, and consecutive images of most positions were separated by the 95 minute orbital period. The sky was fully mapped after 6 months of observations. *WISE* operated from 2010 January 7 to 2011 February 1, completing two maps of the sky and beginning a third one. Images were obtained through all four bands during the first 7 months of the mission, which is known as the 4-Band Cryo phase. Because of the successive exhaustion of its outer and inner cryogen tanks, observations were performed through $W1$, $W2$, and $W3$ from 2010 August 6 to 2010 September 29 (3-Band Cryo) and through $W1$ and $W2$ for the remaining 4 months of the mission (NEOWISE Post-Cryo; Mainzer et al. 2011).

WISE has produced images and source catalogs as its data products (Cutri et al. 2012a, 2012b). The images consist of the single exposures from each of the three phases and coadditions of the single exposures from the 4-Band Cryo and 3-Band Cryo phases, which are known as the Atlas images. Sources detected in the Atlas images are tabulated in the All-Sky Source Catalog and the 3-Band Cryo Source Working Database, respectively. Sources from the single exposures are provided in the All-Sky Single Exposure (L1b) Source Table, the 3-Band Cryo Single Exposure (L1b) Source Table, and the Post-Cryo Single Exposure (L1b) Source Table. In areas of sky that are unconfused and near the ecliptic plane, the images from *WISE* typically achieved a signal-to-noise ratio (S/N) of 5 at $W1 = 16.8$, $W2 = 15.6$, $W3 = 11.3$, and $W4 = 8.0$. The sensitivity was better at higher ecliptic latitudes because of the larger number of exposures. The typical astrometric precision was $0\farcs4$ at S/N = 20.

3. SEARCH STRATEGY

If the Sun has a distant companion, it probably has a spectral type of late T or later given the detection limits from previous all-sky surveys (Section 7). If such an object has been detected by *WISE*, it is likely that it would be found during the photometric searches for T and Y dwarfs in the solar neighborhood (Cushing et al. 2011; Kirkpatrick et al. 2011, 2012). However, it could be missed by those surveys if it has unusual colors. In addition, if a solar companion is in the portion of sky that was mapped twice during the 4-Band Cryo phase, it could have a large enough parallactic motion that it appears elongated or has a degraded S/N in the 4-Band Cryo Atlas images, which are the source of the All-Sky Source Catalog that has been used in the photometric surveys. To avoid these possible obstacles, I have performed a search that is based on parallactic motion instead of colors. This analysis also is capable of uncovering a companion that has been detected in previous all-sky surveys like 2MASS but has been overlooked.

WISE mapped each position in the sky 2–3 times at intervals of ∼6 months. Because of the timing of those observations, they capture nearly the full amplitude of parallactic motion. For distances of 1000–100,000 AU, a solar companion should move $20''$–$0\farcs02$ from orbital motion and $400''$–$4''$ from parallactic motion during a 6 month period between *WISE* observations. Thus, the motion of a companion should be dominated by parallax, and it should be easily detected given the astrometric precision of *WISE* data. Although the goal of this study is the measurement of such parallactic motion, my analysis is also capable of detecting the large proper motions of members of the solar neighborhood, as demonstrated by the discovery of the nearby binary brown dwarf WISE J104915.57−531906.1 (hereafter WISE 1049−5319, Luhman 2013).

4. PROCESSING OF *WISE* DATA

4.1. Catalog of Single- and Multi-epoch Objects

To search for objects that exhibit large motions within the images from *WISE*, I began by retrieving the three single exposure catalogs for the mission described in Section 2. Sources within a few degrees of the ecliptic poles (∼0.05% of the sky) were omitted because the time required to retrieve and process those data would be prohibitive for such a small area of sky. No other areas of the sky were excluded in this study. My analysis in Luhman (2013) excluded single-exposure sources that had 2MASS counterparts within $3''$. That restriction was not imposed in this study. I retained objects that had profile-fit photometry in at least one band (w1sigmpro ≠ null or w2sigmpro ≠ null or w3sigmpro ≠ null or w4sigmpro ≠ null) and that were not flagged as possible artifacts in any band (cc_flag ≠ D, P, R, H, O, L). The resulting catalog contained ∼18.5 billion entries.

Several steps in the following analysis of the catalog from *WISE* involved the matching of sources based on celestial coordinates and other parameters (e.g., observation date). This matching was performed with the programs TOPCAT[1] and STILTS[2] (Taylor 2005, 2006). In the first step, I sought to identify each group of sources from the *WISE* catalog that corresponded to the detections of an object from consecutive scans by the satellite. To do this, I searched for all groups of *WISE* sources in which the closest neighbors were separated by $<1\farcs5$ and <1 day. A member of a group was rejected if it had an accurate photometric measurement ($\sigma < 0.1$ mag) in any band that differed by >1 mag from the median value of the group. A group was rejected if it contained less than five members, contained multiple detections from the same image (indicative of an extended source), or was within the radius of a source in the 2MASS All-Sky Extended Source Catalog, where the radius is represented by the parameter "r_ext" in that catalog. I refer to the period of time spanning one of these groups as an "epoch." Given the interval of 95 minute between successive scans of a given position, the detections of an object moving faster than $\sim1''$ hr^{-1} would not be matched as a group. Mainzer et al. (2011) have developed a method that is capable of identifying sources detected by *WISE* that have such large proper motions, which consist of small bodies in the inner solar system.

To match groups of detections from epochs across the entire duration of the *WISE* survey, I computed the mean coordinates for each group and searched for groups of these mean coordinates with separations of $<1\farcs5$ between neighbors. The resulting groups of detections from different epochs are referred to as "multi-epoch objects." The remaining unmatched groups of detections are referred to as "single-epoch objects." A multi-epoch object was rejected if the median magnitudes of its single-epoch groups differed by >1 mag in all detected bands. For each of the retained multi-epoch objects, I derived proper motions in right ascension and declination by applying linear regression to the astrometry and observation dates of its single-exposure detections. The standard errors associated with

[1] http://www.starlink.ac.uk/topcat/
[2] http://www.starlink.ac.uk/stilts/

these fits were also computed. Because consecutive epochs for a given position are separated by ∼6 months for most areas of the sky, the proper motions for the multi-epoch objects are below $\sim 3''$ yr^{-1}. A star with a higher proper motion appears as two or three unmatched single-epoch objects. The latter category also contains variable sources, spurious detections, and portions of blended objects. In Section 6, I search the single-epoch objects for unmatched epochs of high proper motion stars.

Most of the single- and multi-epoch objects span ∼1 day and ∼6/12 months, respectively. However, because of occasional irregularities in the scanning pattern of *WISE* (e.g., Moon avoidance), some areas of sky were observed at multiple epochs separated by days or weeks (in addition to the epochs spanning ∼6 or 12 months), resulting in a few multi-epoch objects that span $\lesssim$1 month. For the purposes of this study, multi-epoch objects that cover periods of <50 days are treated as single-epoch objects.

Through my matching process, it is possible that a subset of the single-epoch groups for a star are matched while the remainder are not. For instance, consider an object that has a small proper motion but a large parallax ($\pi \gtrsim 0\farcs75$), such as a companion to the Sun. If it appears in the part of the sky that was imaged over a span of a full year, then the two epochs that are separated by a year could be matched while the intervening epoch is unmatched. The matched pair of epochs would exhibit little motion. Objects of this kind should be found through my examination of single-epoch objects in Section 6. On the other hand, if $\pi \lesssim 0\farcs75$, then all three epochs would be matched, and the net motion would again be near zero. No single-epoch group would remain with which to identify such an object. As a result, my analysis is not sensitive to a solar companion with $\pi \lesssim 0\farcs75$ ($d \gtrsim 1.3$ pc). However, it is unlikely that a companion could maintain a stable, bound orbit at that distance (Torbett & Smoluchowski 1984). In another scenario in which only a subset of groups are matched for an object, two consecutive epochs are matched while a third epoch is not matched because of statistical errors in the astrometry of an object near $\sim 3''$ yr^{-1}, or because of the non-uniform motion over the course of a year for an object that has comparable proper and parallactic motions. Once again, I expect to find such objects by examining the unmatched single-epoch groups in Section 6. They also may exhibit significant proper motions through their matched groups (Section 5).

4.2. Comparison to WISE All-Sky Catalog

I have used the *WISE* All-Sky Source Catalog to refine my catalog of multi-epoch and single-epoch objects. I began by identifying pairs of entries from the two catalogs with separations less than $1.5''$. If an object in my catalog had an All-Sky counterpart that was flagged as a possible artifact in any band (cc_flag = D, P, R, H, O, L), then it was rejected. This step removed objects that were flagged as artifacts in the All-Sky Source Catalog but not in the single-exposure catalogs. Because the All-Sky Source Catalog is based on coadditions of the single-exposure images, it should be more reliable than the single-exposure catalogs. Therefore, I also rejected objects that lacked a matching All-Sky source, except for single-epoch objects whose locations were observed more than once during the 4-Band Cryo phase. The latter exception was made because an object moving fast enough so that its detections at different epochs are not matched to each other ($\mu \gtrsim 3''$ yr^{-1}) is likely to have an elongated point-spread function or degraded S/N in the Atlas images, and thus may not be reliably detected during the construction of the All-Sky Source Catalog.

I compared the photometry for matching entries in my catalog and the All-Sky Source Catalog. My mean fluxes are systematically brighter than the All-Sky fluxes for the faintest objects ($S/N_{\text{All-Sky}} \lesssim 10$). This trend is likely caused by the fact that, near the detection limit, a star is detected in only a subset of the individual *WISE* exposures, and those detections are biased to brighter fluxes because upward fluctuations of the background noise facilitate detection. Meanwhile, the All-Sky photometry is measured from the Atlas images, which are constructed from all of the single exposures, not just those with detections. Therefore, for most objects in my catalog, I have adopted the photometry of the All-Sky counterparts. However, for single-epoch objects whose locations were observed at more than one epoch during the 4-Band Cryo phase, I use the mean photometry from the single-exposure sources because the All-Sky photometry is based on all images during 4-Band Cryo phase rather than only the images corresponding to the single epoch in question.

Data from the All-Sky Source Catalog are also useful for assessing the completeness of my catalog and identifying the magnitude limit for my search for high proper motion objects. For each band, I find that the percentage of All-Sky sources that have counterparts in my catalog during the 4-Band Cryo phase is roughly constant as a function of magnitude down to $S/N_{\text{All-Sky}} \sim 10$. The value of this percentage at $S/N_{\text{All-Sky}} \gtrsim 10$ is ∼100% for $W1$ and $W2$, ∼70% for $W3$, and ∼50% for $W4$. Based on visual inspection of a sample of the All-Sky detections in $W3$ and $W4$ that are absent from my catalog, many of them appear to be spurious. Thus, the completeness of my catalog relative to the All-Sky Source Catalog at $W3$ and $W4$ is probably better than these percentages would imply. To select the magnitude limit down to which I will inspect candidate high proper motion objects (Sections 5 and 6), I have examined the number of single-epoch objects as a function of $W2$. The distribution increases rapidly with fainter magnitudes near $W2 \sim 14.5$ ($S/N_{\text{All-Sky}} \sim 14$), which is a reflection of the fact that the probability of detecting an object in multiple epochs decreases as the detection limit is approached. This threshold is a practical limit for the inspection of candidate proper motion sources because examining the large number of candidates at fainter magnitudes would not be feasible. In addition, the All-Sky Source Catalog has a high level of completeness at this $W2$ magnitude; it is 95% complete at S/N ∼ 10 in $W2$, which corresponds to $W2 \gtrsim 14.8$ for 95% of the sky (Cutri et al. 2012b). Therefore, in the next two sections, I will search for high proper motion sources down to $W2 = 14.5$. I do not consider sources that have $W2 > 14.5$ but are detected in other bands because the limits on the mass of a solar companion provided by $W1$ are higher than those from $W2$ (Section 7) and because multi-epoch images in $W3$ and $W4$ are not available for the entire sky.

5. HIGH PROPER MOTION CANDIDATES AMONG MULTI-EPOCH OBJECTS

The analysis in the previous section produced a catalog that contains the mean coordinates and proper motions computed from groups of *WISE* single-exposure sources that are near each other on the sky. Some of these entries are based on groups of detections from multiple epochs, which typically span ∼6 or 12 months, while others are based on groups of detections at single ∼1 day epochs for which a matching group at another epoch was

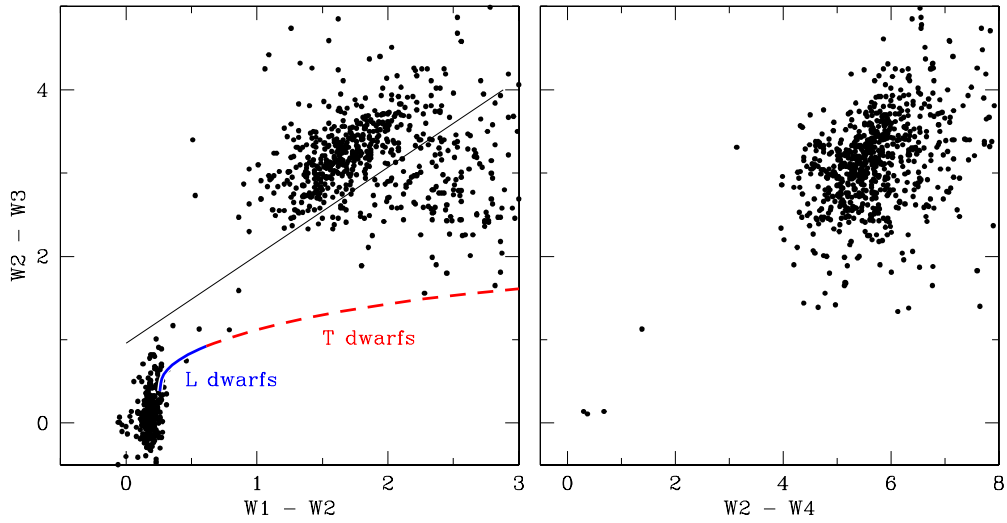


Figure 1. Color–color diagrams for new high proper motion candidates ($\mu/\sigma_\mu > 5$) detected with multi-epoch astrometry from *WISE*. In each diagram, the colors in the group near the origin are consistent with those of main-sequence stars and brown dwarfs; all of these candidates have detections in 2MASS that confirm their large motions. Most of the objects in the redder group are probably galaxies or evolved stars; none have motions confirmed by 2MASS. For reference, the left diagram includes the colors produced by fits to absolute magnitudes of L and T dwarfs (solid and dashed curves, Dupuy & Liu 2012) and a boundary below which Kirkpatrick et al. (2011) selected candidate T and Y dwarfs (solid line). The candidate that is between the two groups in the right diagram is the binary brown dwarf WISE 1049−5319 (Luhman 2013).

(A color version of this figure is available in the online journal.)

not present within 1.″5. In this section, I search for high proper motion objects among the former, which should have motions of $\lesssim$3″ yr^{-1}.

For the initial sample of high proper motion candidates, I selected multi-epoch objects that exhibit $\mu/\sigma_\mu > 5$, have counterparts within 1.″5 in the All-Sky Source Catalog, and are brighter than the threshold of $W2 = 14.5$ that was adopted in the previous section for this survey. To assess the completeness of this sample, I identified the counterparts of previously known high proper motion stars and brown dwarfs (e.g., Subasavage et al. 2005; Lépine & Shara 2005; Faherty et al. 2009) among the multi-epoch objects. This was also done for the sample of single-epoch objects described in the next section, which encompasses stars with $\mu \gtrsim 3''$ yr^{-1}. These two samples recovered 90% of previously known objects with $\mu > 1''$ yr^{-1} and $W2 < 14.5$. Most of the 10% of stars that were not recovered (6.5%) had large astrometric errors due to saturation. The remaining stars were blended with a binary companion or a field star, rejected during construction of the catalog (e.g., cc_flag), or had uncertain proper motions ($\mu/\sigma_\mu < 5$).

After removing known high proper motion stars from the sample, I searched for matches within 3″ in the 2MASS Point Source Catalog for the remaining candidates. If a match was not present, the 2MASS images were inspected for a detection at either the *WISE* position or a location that would be consistent with the proper motion derived from the *WISE* astrometry. A detection in 2MASS at <3″ from the *WISE* coordinates implies a proper motion of $\lesssim$0.″3 yr^{-1}; I rejected most of these candidates since I wish to search for objects with higher proper motions. Because a distant solar companion would have a negligible proper motion, it could appear in 2MASS near the location of the *WISE* detection. Therefore, when rejecting the candidates with <3″ counterparts from 2MASS, I required that $\mu+\sigma_\mu < 4''$ yr^{-1}, which corresponds to $\pi \gtrsim 1''$ for a solar companion. For the remaining unrejected candidates, I inspected the *WISE* Atlas images to remove extended objects, spurious detections, and blends of multiple objects, which can produce false indications of motion between epochs. The sample contained ∼1500 viable candidates at this stage. For 761 of these objects, I found detections in 2MASS that confirmed their high proper motions.

The *WISE* colors of the high proper motion candidates provide constraints on their nature. Spurious detections are fairly common at $W3$ and $W4$ in the *WISE* catalogs, which can lead to misclassifications during an analysis of the colors. Therefore, I inspected the *WISE* Atlas images of the candidates and removed photometric measurements for which detections were not visually apparent. In Figure 1, I plot $W2 - W3$ versus $W1 - W2$ and $W2 - W3$ versus $W2 - W4$ for the candidates that have photometry in these bands. In each diagram, the candidates form two distinct populations, one with neutral and moderately red colors and the other with much redder colors. These groups also correspond to candidates with and without 2MASS counterparts, respectively. The colors in the first group are consistent with those of main-sequence stars and brown dwarfs. Most of the candidates in the second group are probably evolved stars that are projected against extended emission or galaxies, although a few could be very cool brown dwarfs, as illustrated in Figure 1, which shows the boundary below which Kirkpatrick et al. (2011) selected candidate T and Y dwarfs. If a candidate appears above that boundary or resides in the red group of candidates in $W2 - W3$ versus $W2 - W4$, I treat it as having non-stellar colors. All of the candidates that are absent from Figure 1 (no $W3$ or $W4$) are consistent with stellar colors since $W1 - W2$ spans a wide range of values among stars and brown dwarfs.

For the small number of candidates that have stellar colors and lack confirmation of their motions via 2MASS detections, I inspected images from public data archives of various observatories and wide-field surveys. All of these candidates showed evidence of extended emission or blended stars, leading to their rejection from the sample. The remaining viable candidates consist of the 761 objects with motions confirmed by 2MASS. This final sample is presented in Table 1, which contains their designations and photometry from 2MASS and *WISE* and proper

motions measured from the combined astrometry in those surveys. In Luhman (2013), I reported the discovery of one member of this sample, WISE 1049−5319, which is a binary system whose components are near the L/T transition.

To illustrate how the candidates are distributed in magnitude and proper motion, I have plotted them in diagrams of $W2$ versus proper motion in Figure 2. These proper motions are based on the *WISE* data alone. The left panel of Figure 2 shows the sample of candidates prior to the analysis of the color–color diagrams, and the right panel shows the final sample of 761 objects. In addition to those multi-epoch candidates, a high proper motion object found through inspection of the single-epoch objects in Section 6 is also included in Figure 2 and Table 1. It is the candidate with the largest proper motion ($\mu \sim 5''$ yr^{-1}). The parallactic motion of a solar companion within 1 pc would appear as a motion of $\gtrsim 4''$ yr^{-1} in these data. None of the multi-epoch candidates in Figure 2 exhibit a motion of this size, which was expected given the matching threshold of $1\farcs5$ for those candidates. The object from the single-epoch analysis does exceed $4''$ yr^{-1}; the question of whether it could be a solar companion is discussed in Section 6.

One of the brightest candidates in Figure 2 is WISE 1049−5319 ($W2 = 7.3$). It is plotted in Figure 2 with a motion of $1\farcs3$ yr^{-1}, which was computed from two epochs of astrometry. However, its parallactic and proper motions partially canceled each other between those epochs, and its true proper motion is larger than that value ($2\farcs8$ yr^{-1}, Luhman 2013). A third epoch of data is available for WISE 1049−5319, but it was not matched to the other pair because the separation was greater than $1\farcs5$, and hence it appears as a single-epoch object in my catalog.

Table 1
New High Proper Motion Objects

Column Label	Description
WISE	Source name from the *WISE* All-Sky Source Catalog
2MASS	Source name from the 2MASS Point Source Catalog
muRA	Proper motion in right ascension
e_muRA	Error in muRA
muDe	Proper motion in declination
e_muDE	Error in muDE
Jmag	2MASS J magnitude
e_Jmag	Error in Jmag
Hmag	2MASS H magnitude
e_Hmag	Error in Hmag
Ksmag	2MASS K_s magnitude
e_Ksmag	Error in $K_s$mag
W1mag	*WISE* $W1$ band magnitude
e_W1mag	Error in $W1$mag
W2mag	*WISE* $W2$ band magnitude
e_W2mag	Error in $W2$mag
W3mag	*WISE* $W3$ band magnitude
e_W3mag	Error in $W3$mag
W4mag	*WISE* $W4$ band magnitude
e_W4mag	Error in $W4$mag

Notes. J, H, and K_s are from the 2MASS Survey Point Source Reject Table for WISE J051505.56−434333.9, WISE J175839.20−583931.6, and WISE J191453−181044 and from the 2MASS Point Source Catalog for all other 2MASS sources. $W1 - W4$ are from the *WISE* All-Sky Source Catalog; detections that appear false or unreliable based on visual inspection have been omitted. Proper motions are based on astrometry from 2MASS (if available) and *WISE*. For WISE 1049−5319, additional measurements of astrometry were included in the proper motion measurement (Luhman 2013).

(This table is available in its entirety in a machine-readable form in the online journal. A portion is shown here for guidance regarding its form and content.)

6. HIGH PROPER MOTION CANDIDATES AMONG SINGLE-EPOCH OBJECTS

The single-epoch objects should contain stars with motions of $\mu \gtrsim 3''$ yr^{-1}, which corresponds to parallactic motions at distances of $\lesssim$1.3 pc. Thus, a solar companion should appear in this sample if it was detected by *WISE*. As in the analysis of the multi-epoch objects, I adopted a limit of $W2 \leqslant 14.5$ for the single-epoch sample, which was satisfied by 4.4 million objects. Many of these sources are either spurious detections or stars that appeared at the same position in the images from different epochs but that have only one epoch in the catalog. The latter typically consist of blends of multiple sources (e.g., binary stars) and stars near the faint limit of the sample. In an attempt to remove these contaminants, I rejected single-epoch objects that have counterparts within $3''$ in 2MASS. If a substellar companion to the Sun was detected by 2MASS, it could have been overlooked by surveys for nearby brown dwarfs with 2MASS photometry if its S/N was low. Therefore, I have retained single-epoch objects with 2MASS counterparts if they have S/N < 10 in all 2MASS bands. As an additional filter against stationary stars that are missing epochs in the catalog, I rejected single-epoch objects that had more than one counterpart within $1''$ at a different epoch in the single-exposure catalogs. In other words, a stationary star may have single-exposure detections at all epochs, but in some epochs the number of detections is too low to result in an entry in my catalog (see Section 4.1), which can occur for components of blended stars. A single-epoch object was rejected in this way only if no other single-epoch objects were within $5''$ since such a source could be responsible for a matching single-exposure detection at the other epoch. Meanwhile, I also removed single-epoch objects with $<1''$ single-exposure matches at a different epoch if multi-epoch objects were present within $5''$ and at the same epoch as the former, which is indicative of a pair of blended stars. The sample contained ∼228,000 objects after these criteria were applied. If a solar companion or other rapidly moving source is in the initial sample of single-epoch objects, then there is a ∼1.6% probability that it would be rejected through these criteria based on their application to a sample of random positions across the sky that are not within $1\farcs5$ from an All-Sky source (i.e., positions that do not coincide with multi-epoch objects).

Because the entire sky was mapped during the 4-Band Cryo phase, a solar companion should appear in the data from that period if it exists and is detectable by *WISE*. My sample contains ∼61,000 single-epoch objects from that phase. The size of this sample could have been reduced by a factor of >100 by excluding objects that are not sufficiently red in the *WISE* bands to be brown dwarfs, but that was not done here so that the survey is unbiased in terms of colors. To check whether each candidate is a high proper motion object, I visually inspected all epochs of *WISE* images at its location. For epochs during the 4-Band Cryo phase, I used the Atlas images if a candidate's position was observed only once. If it was observed at two epochs spanning ∼6 months, the single-exposure images at each epoch were coadded separately. I employed the Atlas images for the 3-Band Cryo phase and the coadded single-exposure images for the Post-Cryo phase.

In Section 4.1, I described two scenarios in which only a subset of epochs of a moving star might be matched to form a multi-epoch object. In one case, two epochs of a solar companion that span a year are matched, but the intervening

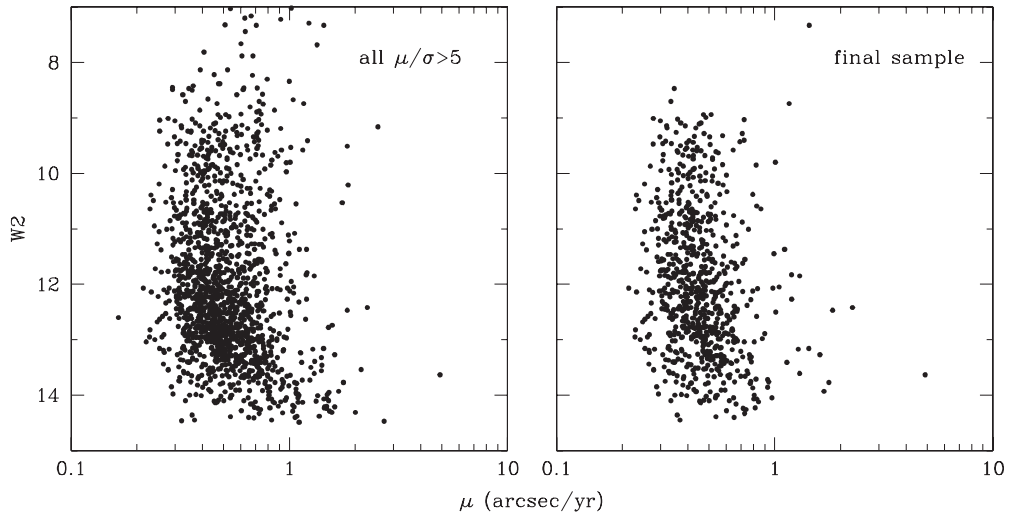


Figure 2. Left: $W2$ vs. proper motion for new high proper motion candidates ($\mu/\sigma_\mu > 5$) detected with multi-epoch astrometry from *WISE*. Right: the final sample of candidates (Table 1), consisting of the candidates that have stellar colors in the *WISE* bands (Figure 1) and that are not rejected as motionless stars by other available data. All but one of these objects are detected by 2MASS. A companion to the Sun at a distance of <1 pc would exhibit a proper motion of $\gtrsim$4″ yr^{-1} in these data. One object is in that range, but its motion is inconsistent with the purely parallactic motion expected for a solar companion.

epoch is not. It is also possible that only 2/3 epochs are matched for high proper motion stars, although the two epochs must be consecutive. To search for these kinds of objects, I identified matches between single- and multi-epoch objects that had complimentary, non-overlapping epochs and had $\Delta W2 < 1$ mag. Their images were inspected in the same manner as for the larger sample of singe-epoch objects.

Most of the single-epoch objects in the inspected sample are spurious detections, blends of neighboring stars, and objects that appear in the images at all epochs but have detections in the catalog at only one epoch (some of which are variable sources). The spurious detections are heavily concentrated in areas with high densities of stars or extended emission like the galactic plane and nearby galaxies. I recovered more than 100 previously known high proper motion stars and found two new objects with large motions, WISE 1049−5319 and WISE J085510.83−071442.5. The first epoch of WISE 1049−5319 is a single-epoch object because it was not matched to its other two epochs during construction of the multi-epoch objects, as mentioned in the previous section. WISE J085510.83−071442.5 consists of two single-epoch objects that are separated by 2.″5 and six months. The direction of its motion is inconsistent with purely parallactic motion, and hence it cannot be a solar companion. The proper motion and $W2$ measurement for WISE J085510.83−071442.5 are included in Figure 2. The characteristics of this object will be discussed in more detail in a separate study.

7. CONSTRAINTS ON A SOLAR COMPANION

Among the *WISE* bands, $W2$ provides the best constraints on the presence of a companion to the Sun. Given the typical sensitivities achieved by *WISE* (Section 2), $W2$ is capable of detecting objects at larger distances than $W1$, $W3$, and $W4$ for $W1 - W2 > 1.2$, $W2 - W3 < 4.3$, and $W2 - W4 < 7.6$, respectively. Although $W1$ offers higher S/N at $W1 - W2 < 1.2$, any brown dwarf with that color ($\lesssim$T2) would be easily detected at $W2$ if it is within the range of plausible distances for a solar companion (<1 pc) based on the absolute magnitudes of L and T dwarfs (Dupuy & Liu 2012). The temperatures and absolute magnitudes corresponding to $W2 - W3 \sim 4.3$ and $W2 - W4 \sim 7.6$ have not been empirically determined for brown dwarfs because sufficiently cold objects have not been found, but models of brown dwarfs and giant planets can provide some guidance. The models of brown dwarfs from Burrows et al. (1997, 2003) extend down to 1 $M_{\rm Jup}$ for an age of 5 Gyr ($T \gtrsim 100$ K). The predicted spectra from those models do not exceed $W2 - W3 \sim 4.4$ and $W2 - W4 \sim 7.6$, which would suggest that $W2$ is the most sensitive band for that range of masses. However, models of Jupiter and Saturn from Fortney et al. (2011) computed for no solar irradiation (J. Fortney 2013, private communication) do exhibit colors that are redder than those values. As a result, the $W3$ and $W4$ bands may reach a solar companion at $\lesssim$1–2 $M_{\rm Jup}$ out to greater distances than $W2$, although only ~20% of the sky was mapped twice in those bands at their full sensitivity, which is necessary for a survey based on parallactic motion.

In Figure 3, I show the masses as a function of distance that correspond to my survey limit of $W2 = 14.5$ based on the theoretical spectra from Burrows et al. (1997, 2003) and Fortney et al. (2011). The 1 $M_{\rm Jup}$ brown dwarf at 5 Gyr from Burrows et al. (1997) ($M_{W2} = 24.0$) is fainter than the unirradiated analog of Jupiter from Fortney et al. (2011) ($M_{W2} = 21.5$), resulting in different distance limits at that mass from the two models. I have also included constraints from 2MASS, *IRAS*, and reflected light surveys in Figure 3. The limits for 2MASS and *IRAS* are derived with $J = 16.6$ and $F_\nu(12\,\mu{\rm m}) = 0.4$ mJy (S/N $\gtrsim$ 5) and the brown dwarf models from Burrows et al. (2003). For reflected light, I show the typical limit for current surveys ($R = 21$; e.g., Sheppard et al. 2011) and the magnitude that should be reached by future facilities like the Large Synoptic Survey Telescope ($R = 26$). For the reflected light limits, I adopted the average V magnitudes of Jupiter, Saturn (ringless), and Neptune and assumed that each planet has $V - R = 0$. Because brown dwarfs and gas giants are expected to have similar radii at masses above ~1 $M_{\rm Jup}$, the reflected light limits for Jupiter have been adopted for all higher masses in Figure 3.

8. DISCUSSION

My analysis of the multi-epoch astrometry from *WISE* has uncovered several hundred new high proper motion objects,

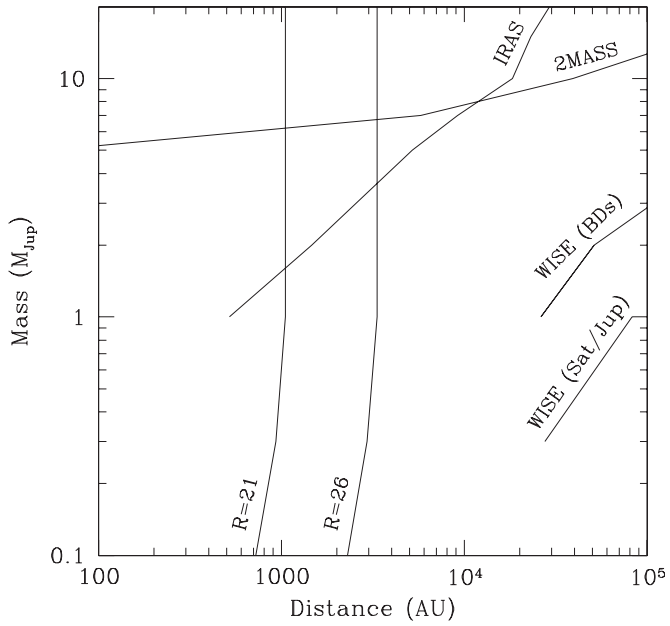


Figure 3. Detection limits for a companion to the Sun produced by current ($R = 21$) and future ($R = 26$) optical surveys and IR images from *IRAS* ($F_{12} = 0.4$ mJy), 2MASS ($J = 16.6$), and *WISE* ($W2 = 14.5$). For the optical limits, objects with masses of $\geqslant 1\ M_{\rm Jup}$ are assumed to have the same fluxes of reflected light because of their similar radii. The IR limits have been derived with fluxes predicted by models of brown dwarfs at an age of 5 Gyr (Burrows et al. 1997, 2003) and (for *WISE* only) models of Saturn and Jupiter in the absence of solar irradiation (Fortney et al. 2011).

including the third-closest known system to the Sun (Luhman 2013). However, none of these sources exhibit a motion that is consistent with a distant companion to the Sun ($d \lesssim 1$ pc). The limit of $W2 = 14.5$ for this survey corresponds to analogs of Saturn and Jupiter at 28,000 and 82,000 AU, respectively, based on the fluxes predicted by models of these planets in the absence of irradiation (Fortney et al. 2011). In their model of a 1 $M_{\rm Jup}$ brown dwarf at an age of 5 Gyr, Burrows et al. (1997) predict fainter mid-IR fluxes than the model of Jupiter, resulting in a closer distance limit of 26,000 AU for that mass. These constraints on the mass of a solar companion at a given distance are an order of magnitude lower than those available from previous IR all-sky surveys, and they indicate that the outer solar system probably does not contain a brown dwarf or a large gas giant planet.

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